



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Marcelo Silva
August 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology Summary

- Data collection via the SpaceX REST API
- Web scraping
- Data wrangling
- Exploratory analysis with SQL
- Exploratory data analysis with visualization
- Interactive visual analysis with Folium
- Dashboards with Plotly Dash
- Predictive modeling (machine learning)

Results Summary

- Success rate trend over time
- Best-performing launch site
- Orbits with a 100% success rate
- Geographic advantage of the launch bases
- Accuracy of the machine learning models
- Best overall performing model
- Most influential factors for success

Introduction

SpaceX

- Space Exploration Technologies Corp. (SpaceX) is a private aerospace company founded in 2002 in the United States by Elon Musk, built around reusable rocket technology.
- Under the traditional approach the rocket was discarded after a single use, pushing costs to roughly US\$165 million per launch.
- SpaceX learned to land the Falcon 9 first stage vertically on land or on drone ships, cutting the cost to roughly US\$62 million.

Project Inputs

- Public launch data gathered through web scraping and the SpaceX REST API, explored and visualized with SQL and Python, then used to build machine learning prediction models.

Project Objectives

- Predict whether the first stage will land successfully
- Estimate launch costs
- Gain competitive advantage
- Identify patterns behind successful landings
- Build a reliable predictive model

Key Questions

- Does payload mass affect the landing?
- Does the target orbit change the outcome?
- Does the launch site make a difference?
- Is landing on ground safer than at sea?
- Does accumulated experience improve results?
- Which algorithm is the most accurate?

Section 1

Methodology

Methodology

- Executive Summary
- Data Collection
 - Historical launch records pulled from the SpaceX REST API and complemented with web scraping from Wikipedia.
- Data Wrangling
 - Falcon 9 launches filtered, missing payload values treated, and a binary landing outcome variable created.
- Exploratory Analysis with SQL
 - Queries run in Jupyter notebooks to surface success rates by site, orbit and year.
- Visual Exploratory Analysis
 - Bar, line and scatter charts built with matplotlib, seaborn and plotly to expose trends and correlations.
- Geospatial Analysis
 - Launch bases mapped with Folium to assess the influence of location on landing outcomes.
- Interactive Dashboard
 - A Plotly Dash application built to filter results by site, orbit and payload range.
- Predictive Modeling
 - Four classification algorithms trained, tuned and compared with scikit-learn

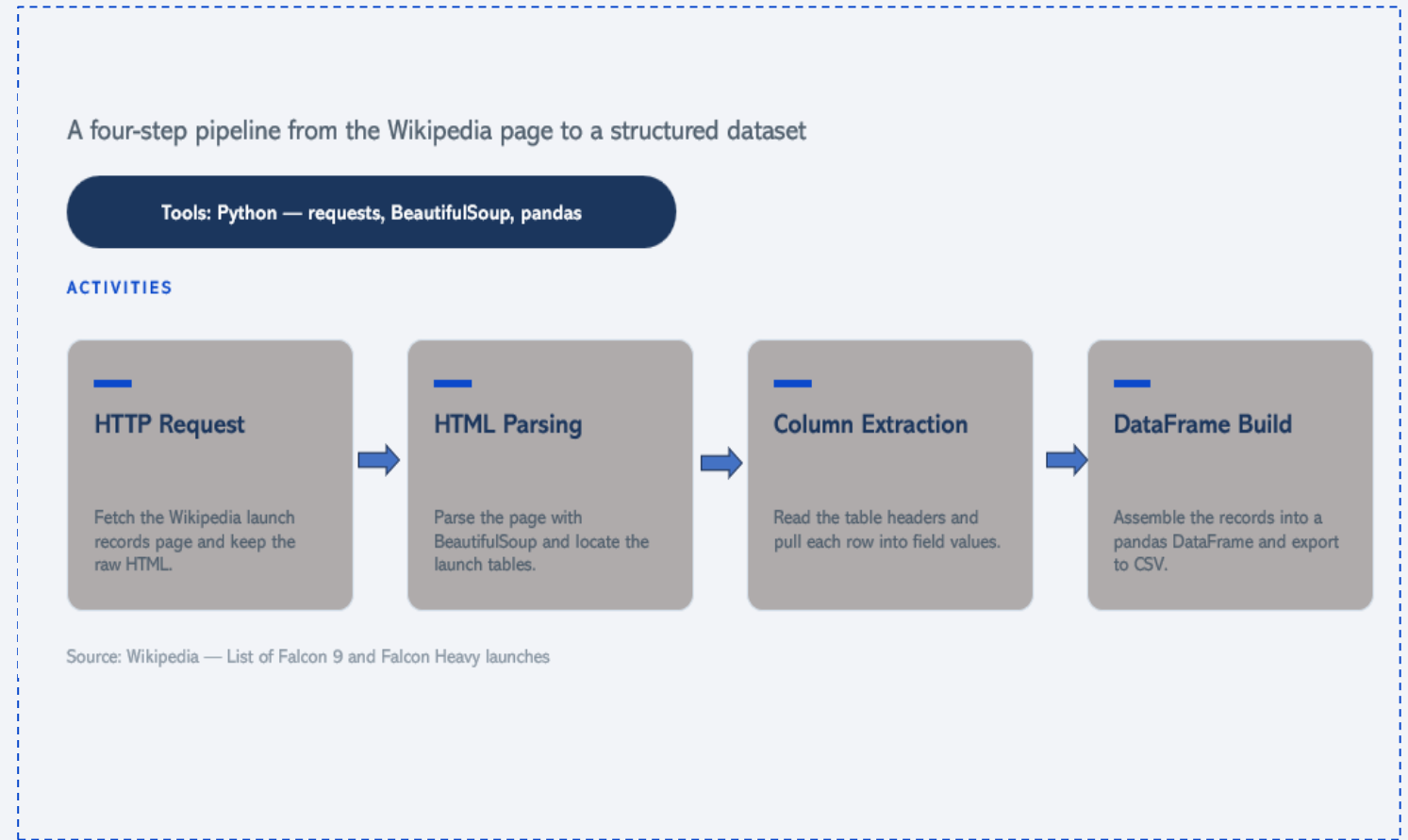
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- GitHub URL of the completed SpaceX API calls notebook:
[capstone spacex/notebooks/01 SpaceX Data Collection API.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/API.ipynb) at main · marcelocards/capstone spacex



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- GitHub URL of the completed web scraping notebook: [capstone spacex/notebooks/02 Spacex data Collection webscraping.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/notebooks/02%20Spacex%20data%20Collection%20webscraping.ipynb) at main · marcelocards/capstone spacex



Data Wrangling

Cleaning and shaping the launch records into a model-ready dataset



Source: SpaceX REST API

- GitHub URL of completed data wrangling : [capstone spacex/notebooks/03 SpaceX Data Wrangling.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/notebooks/03_SpaceX_Data_Wrangling.ipynb) at [main · marcelocards/capstone spacex](https://github.com/marcelocards/capstone_spacex)

EDA with Data Visualization

Querying the launch dataset directly to surface early patterns

Tool: SQL queries run in Jupyter notebooks

ACTIVITIES

Success Rate by Site

Compare landing success across each launch site.

Success Rate by Orbit

Rank orbit types by mission outcome.

Payload Mass by Outcome

Average payload mass for every result class.

Landings over Time

Track how success evolved year by year.

Source: SpaceX REST API

- GitHub URL of your completed EDA with data visualization:
[capstone_spacex/notebooks/05_SpaceX_EDA_Visualization.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/notebooks/05_SpaceX_EDA_Visualization.ipynb) at main · marcelocards/capstone_spacex

EDA with SQL

From loading the table to the specific queries that answered each question

Tools: SQL magic in Jupyter, SQLite over the launch dataset

ACTIVITIES

Load and Inspect

```
SELECT DISTINCT Launch_Site FROM  
SPACEXTBL;
```

Filter and Aggregate

```
SELECT AVG(PAYLOAD_MASS__KG_)  
FROM SPACEXTBL WHERE Customer  
= 'NASA';
```

Group and Rank

```
SELECT Landing_Outcome, COUNT(*)  
FROM SPACEXTBL GROUP BY  
Landing_Outcome;
```

Dates and Extremes

```
SELECT MIN(Date) FROM SPACEXTBL  
WHERE Landing_Outcome LIKE  
'Success%';
```

Source: SpaceX REST API

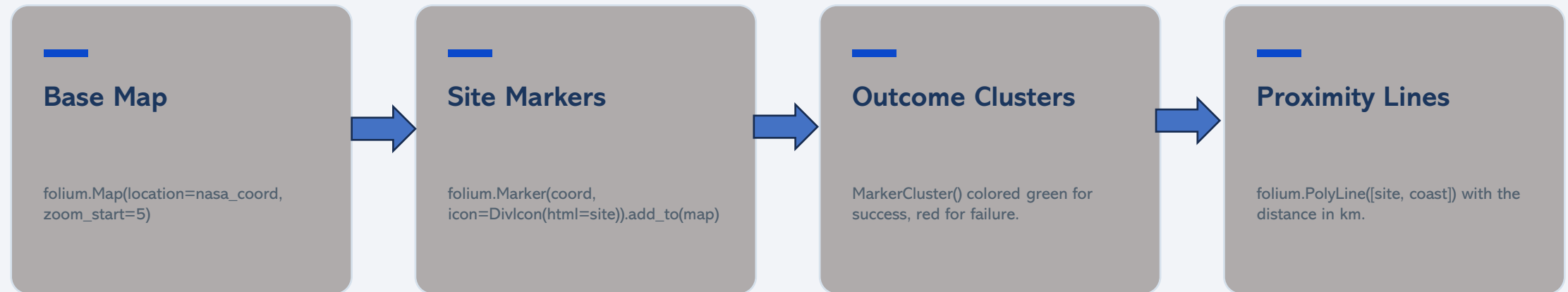
GitHub of URL of completed EDA with SQL notebook:
[capstone spacex/notebooks/04 SpaceX EDA SQL.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/notebooks/04_SpaceX_EDA_SQL.ipynb) at main ·
[marcelocards/capstone spacex](https://github.com/marcelocards/capstone_spacex)

Build an Interactive Map with Folium

From the base map to the markers and distance lines drawn on it

Tools: Folium in Jupyter — Marker, MarkerCluster, PolyLine

ACTIVITIES



Source: SpaceX REST API

Github URL :

[capstone_spacex/notebooks/06_SpaceX_Interactive_Visual_Analytics_Folium.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/notebooks/06_SpaceX_Interactive_Visual_Analytics_Folium.ipynb) at main · marcelocards/capstone_spacex

Build a Dashboard with Plotly Dash

From the app layout to the callbacks that drive each chart

Tools: Plotly Express and Dash — Dropdown, RangeSlider, callbacks

ACTIVITIES

App Layout

```
app.layout = html.Div([dropdown, pie,
slider, scatter])
```

Site Dropdown

```
dcc.Dropdown(id='site-dropdown',
options=sites)
```

Success Pie Chart

```
px.pie(df, values='class',
names='Launch Site')
```

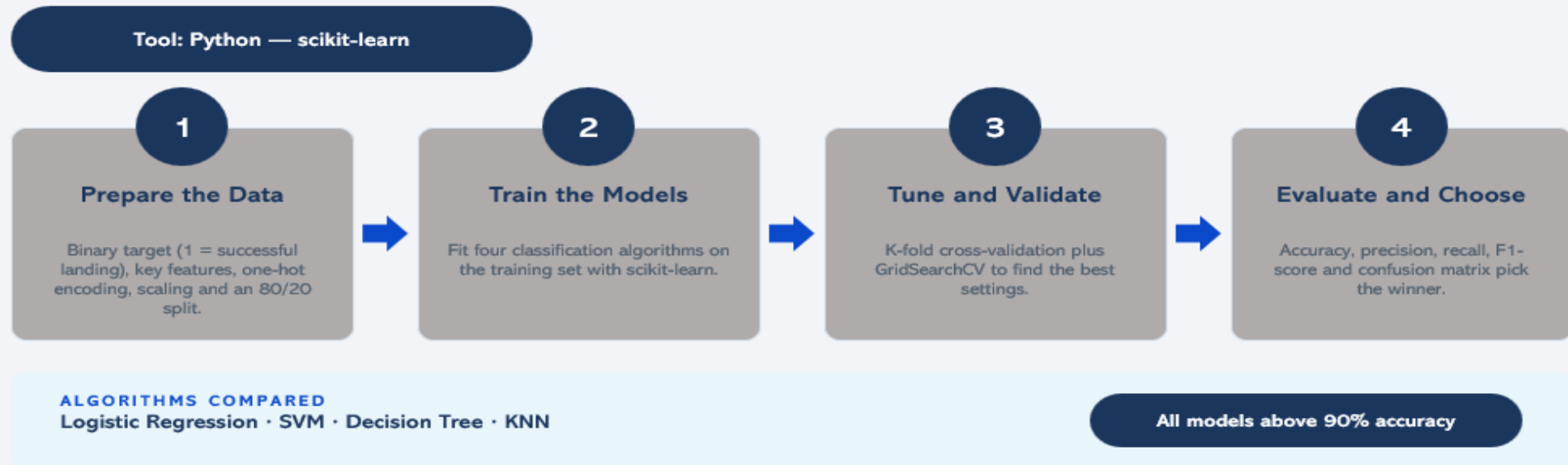
Payload Slider

```
dcc.RangeSlider(0, 10000) feeding a
px.scatter callback.
```

GitHub URL of completed Plotly Dash: [capstone spacex/dash app at main · marcelocards/capstone spacex](https://github.com/marcelocards/capstone_spacex/tree/main)

Predictive Analysis (Classification)

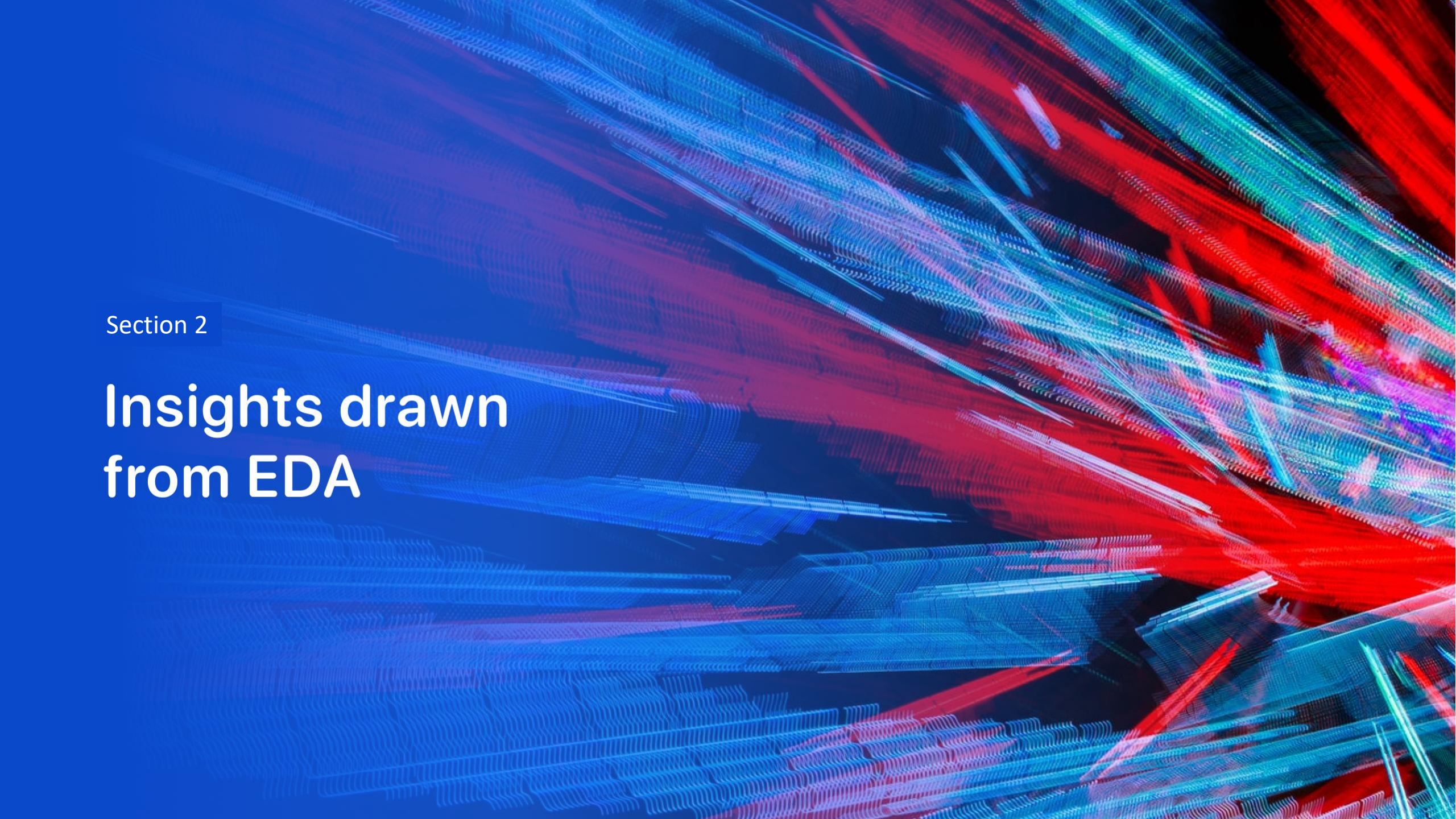
From prepared features to the best-performing landing classifier



- GitHub URL of completed predictive analysis lab:
[capstone_spacex/notebooks/07 SpaceX Machine Learning Prediction.ipynb](https://github.com/marcelocards/capstone_spacex/blob/main/nb.ipynb) at main · marcelocards/capstone_spacex

Results

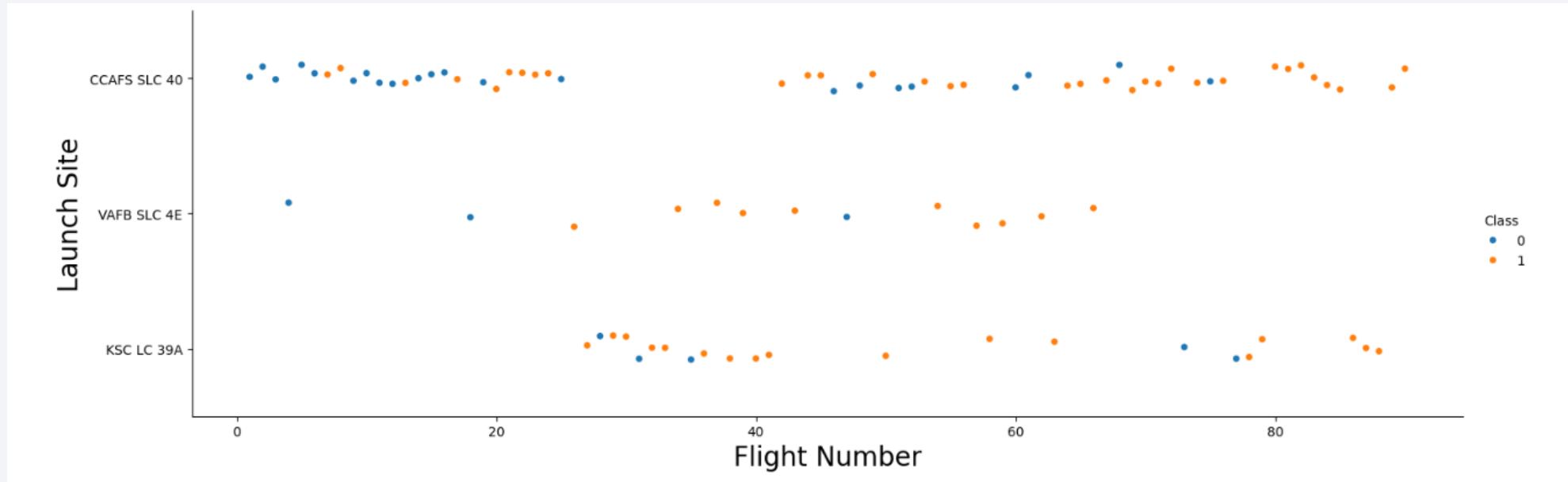
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

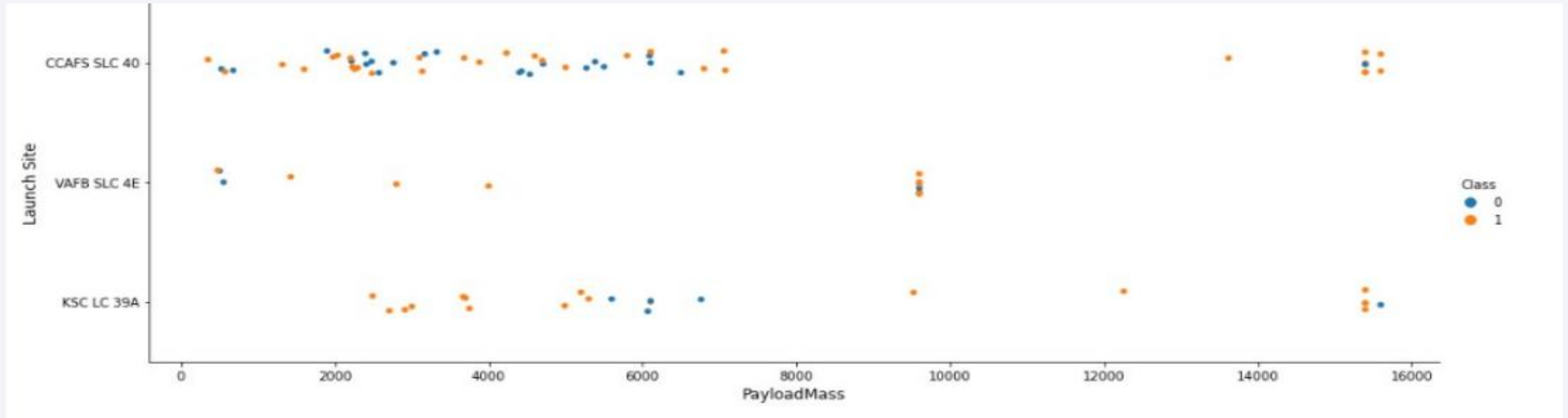
Insights drawn from EDA

Flight Number vs. Launch Site



- Over time, the success rate increased for each launch site—particularly for CCAFS SLC-40, which is the primary launch site (in terms of volume).
- VAFB SLC 4E - high successful rate that represents 1/3 of the total launches

Payload vs. Launch Site



- At VAFB SLC, no rockets are launched with heavy payloads (exceeding 10,000 kg).
- At KSC LC, no rockets were launched for low payloads (less than 2,500 kg).
- At CACAFC CLC, rockets with payloads of less than 7,500 kg and rockets with payloads of more than 13,000 kg were launched; however, there are no rockets with payloads between 7,500 kg and 13,000 kg.

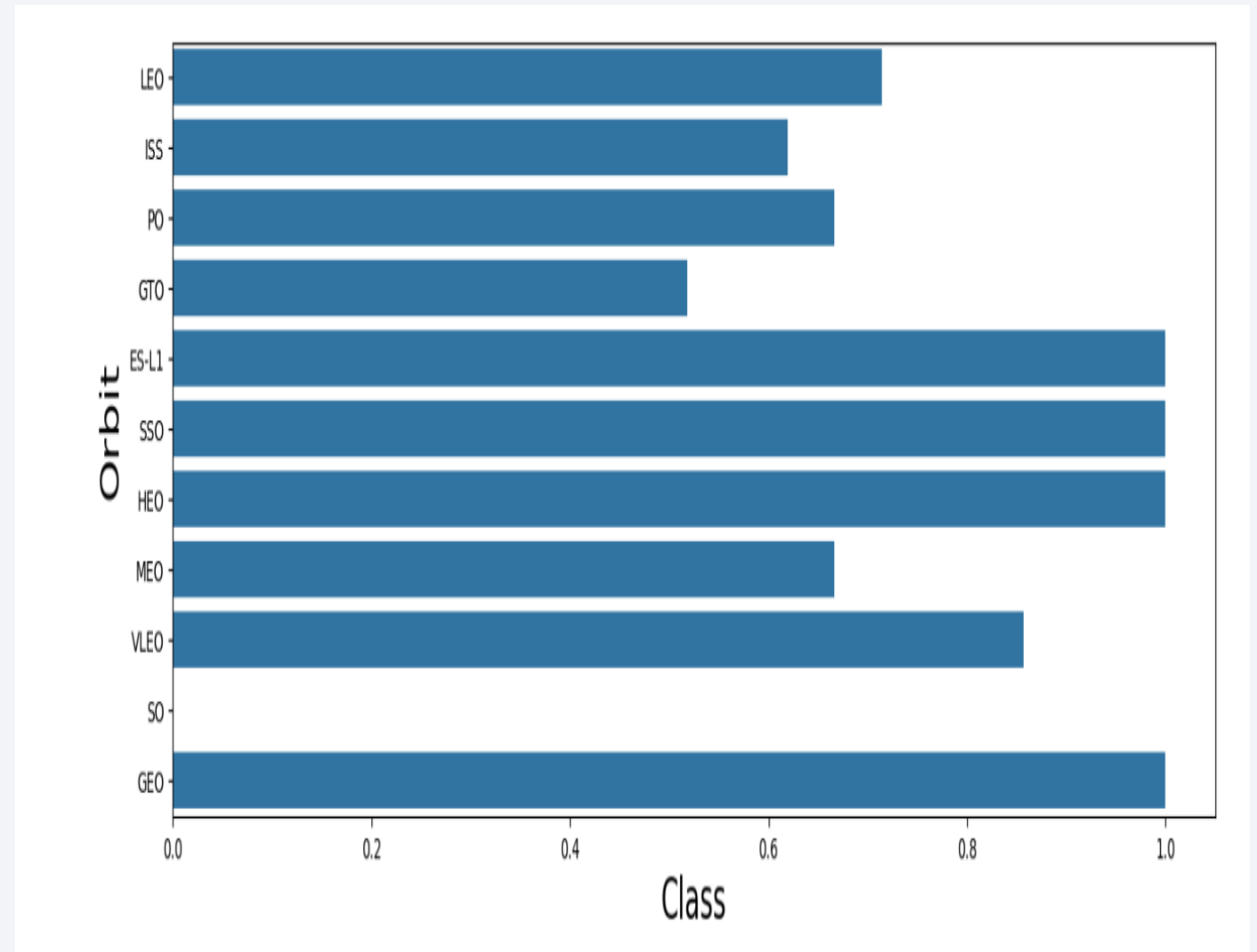
Success Rate vs. Orbit Type

Top performers (Highest): GEO (1.00), SSO (0.99), and HEO (0.99) orbits hold the best scores, closely followed by ES-L1 (0.98).

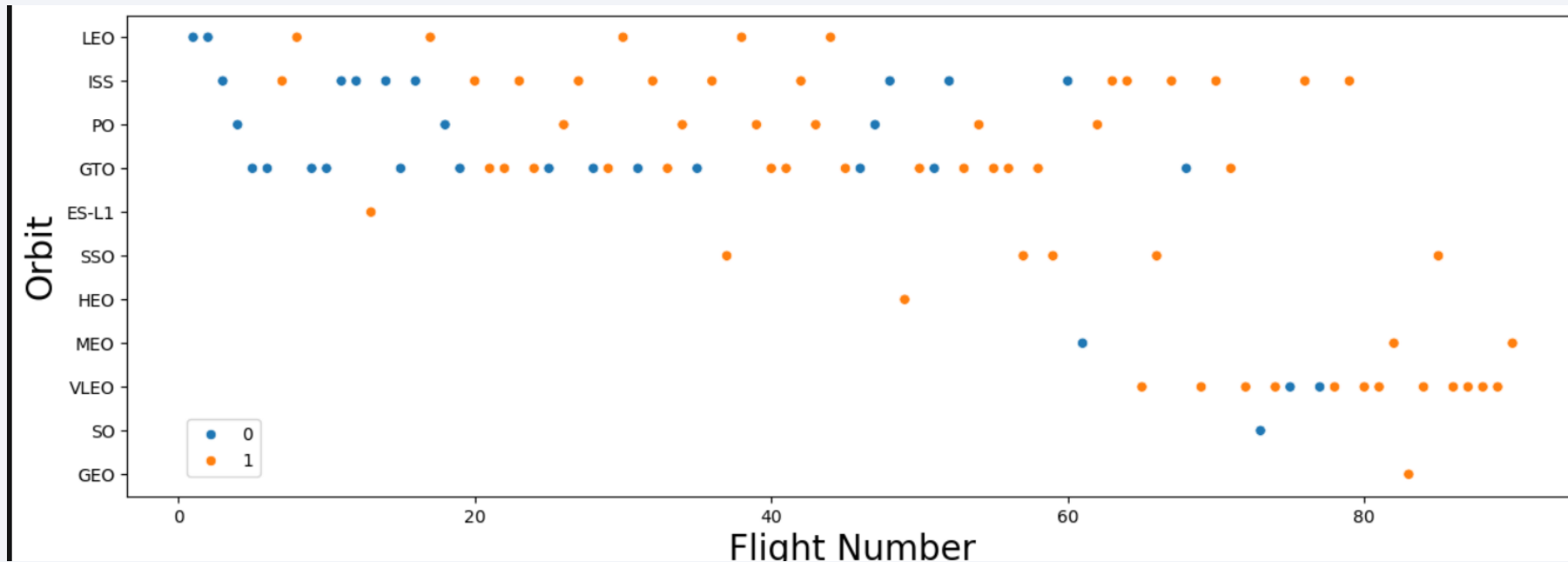
Intermediate performance: SO (0.90) and VLEO (0.88) follow, while the ISS (0.65) and PO (0.70) show moderate scores.

Lowest performance: The GTO orbit (0.55) recorded the lowest score among all analyzed classes.

There is significant variation in the values (ranging from 0.55 to 1.00). Geostationary (GEO) and high-altitude/heliostationary orbits stand out prominently, whereas transfer orbits (GTO) perform significantly worse.

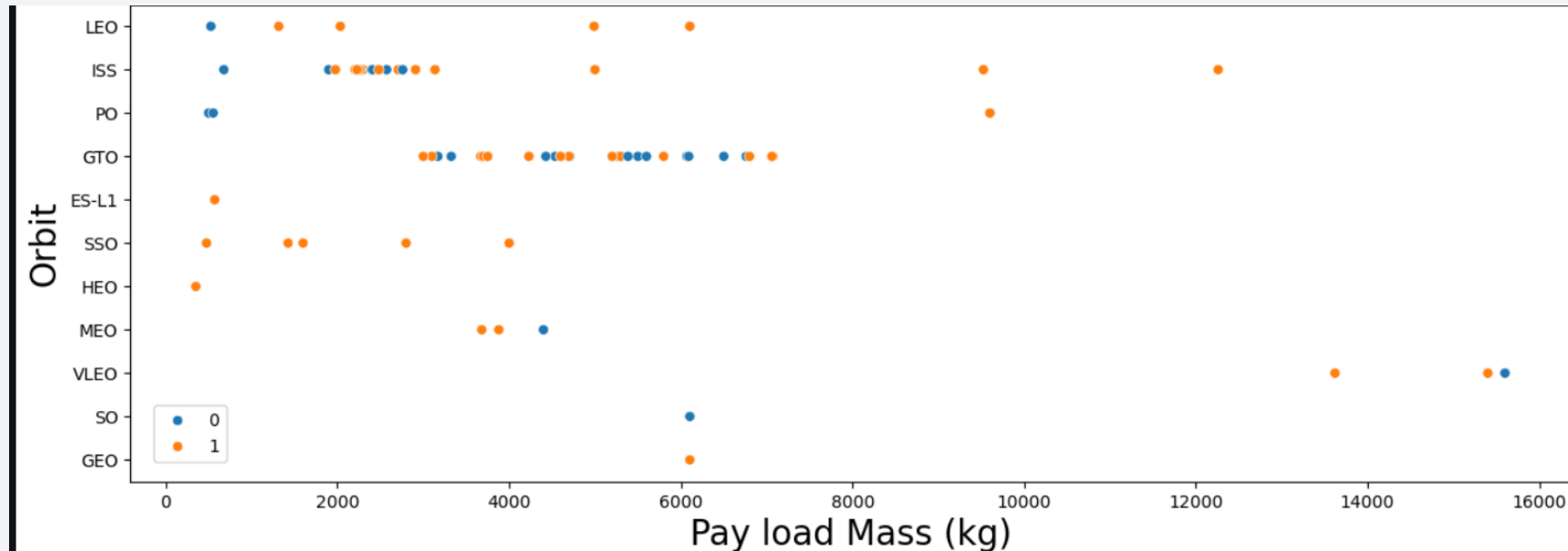


Flight Number vs. Orbit Type



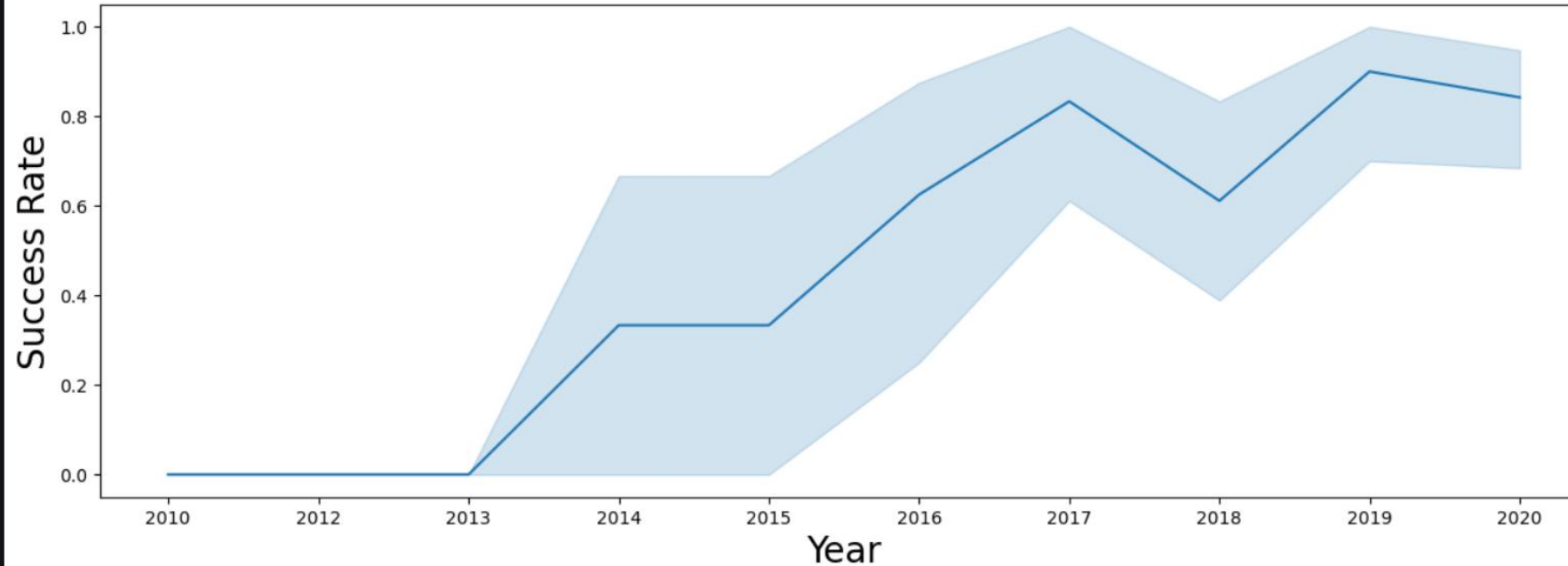
- As expected, there is a higher number of failures during the first 40 launches, with performance improving thereafter.
- GTO and ISS have the highest failure rates.
- The highest success rates occur in orbits with the fewest launches (VLO and SSO).

Payload vs. Orbit Type



- LEO and ISS missions show higher success rates for heavier payloads; however, for GTO, a clear success relationship cannot be defined due to mixed results.
- A total payload limit of 7,500 kg is observed, as there have been few launches exceeding this figure (approximately six).

Launch Success Yearly Trend



Trends over Time

Line charts confirm a continuous improvement in landing success rate across the years.

All Launch Site Names

- I have used : SELECT DISTINCT Launch_Site from SPACEXTABLE
- The four unique launch sites in the space mission

Display the names of the unique launch sites in the space mission

```
%%sql
```

```
SELECT DISTINCT Launch_Site from SPACEXTABLE;
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The query contain: WHERE, LIKE , LIMIT

%%sql

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Total Payload Mass

```
%%sql  
  
SELECT SUM(PAYLOAD_MASS_KG_) AS TOTAL_PAYLOAD FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';  
  
* sqlite:///my\_data1.db  
Done.  
  


| TOTAL_PAYLOAD |
|---------------|
| 45596         |


```

Total payload Mass, using SUM function and WHERE clause.

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%%sql
```

```
SELECT AVG(PAYLOAD_MASS_KG_) AS AVG_PAYLOAD_MASS FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
AVG_PAYLOAD_MASS
```

```
2534.6666666666665
```

Average Payload Mass , using AVE fubction and WHERE

First Successful Ground Landing Date

```
%%sql

SELECT MIN(Date) as LaunchDate FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';

* sqlite:///my_data1.db
Done.

LaunchDate
2015-12-22
```

First successful Ground landing, was achieved using MIN function

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
```

```
SELECT Booster_Version, PAYLOAD_MASS_KG_ FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Booster_Version	PAYLOAD_MASS_KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000, combining WHERE clause with AND operator

Total Number of Successful and Failure Mission Outcomes

```
%%sql

SELECT CASE
    WHEN Mission_Outcome LIKE 'Success%' THEN 'Success'
    WHEN Mission_Outcome LIKE 'Failure%' THEN 'Failure'
END as Mission_Status,
COUNT(*)
FROM SPACEXTABLE
GROUP BY Mission_Status;
```

* [sqlite:///my_data1.db](#)
Done.

Mission_Status	COUNT(*)
Failure	1
Success	100

List the total number of successful and failure mission outcomes – combination of COUNT with GROUP BY

Boosters Carried Maximum Payload

```
SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_  
FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)  
ORDER BY Booster_Version;
```

* [sqlite:///my_data1.db](#)

Done.

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.4	15600
F9 B5 B1049.5	15600
F9 B5 B1049.7	15600
F9 B5 B1051.3	15600
F9 B5 B1051.4	15600
F9 B5 B1051.6	15600
F9 B5 B1056.4	15600
F9 B5 B1058.3	15600
F9 B5 B1060.2	15600
F9 B5 B1060.3	15600

```
SELECT DISTINCT Booster_Version,  
PAYLOAD_MASS__KG_  
FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ =  
(SELECT MAX(PAYLOAD_MASS__KG_)  
FROM SPACEXTABLE)  
ORDER BY Booster_Version;  
Using a subquery
```

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site	Date
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	2015-01-10
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40	2015-04-14

There were speeches in January and February, both landing on drone ship, both being launched from CCAFS - LC 40, Query:

```
SELECT
CASE strftime('%m', Date)
WHEN '01' THEN 'January'
WHEN '02' THEN 'February'
WHEN '03' THEN 'March'
WHEN '04' THEN 'April'
WHEN '05' THEN 'May'
WHEN '06' THEN 'June'
WHEN '07' THEN 'July'
WHEN '08' THEN 'August'
WHEN '09' THEN 'September'
WHEN '10' THEN 'October'
WHEN '11' THEN 'November'
WHEN '12' THEN 'December'
END as Month,
Landing_Outcome, Booster_Version, Launch_Site, Date
FROM SPACEXTABLE
WHERE strftime('%Y', Date) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT Landing_Outcome, COUNT(*) as Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
```

* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

The count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

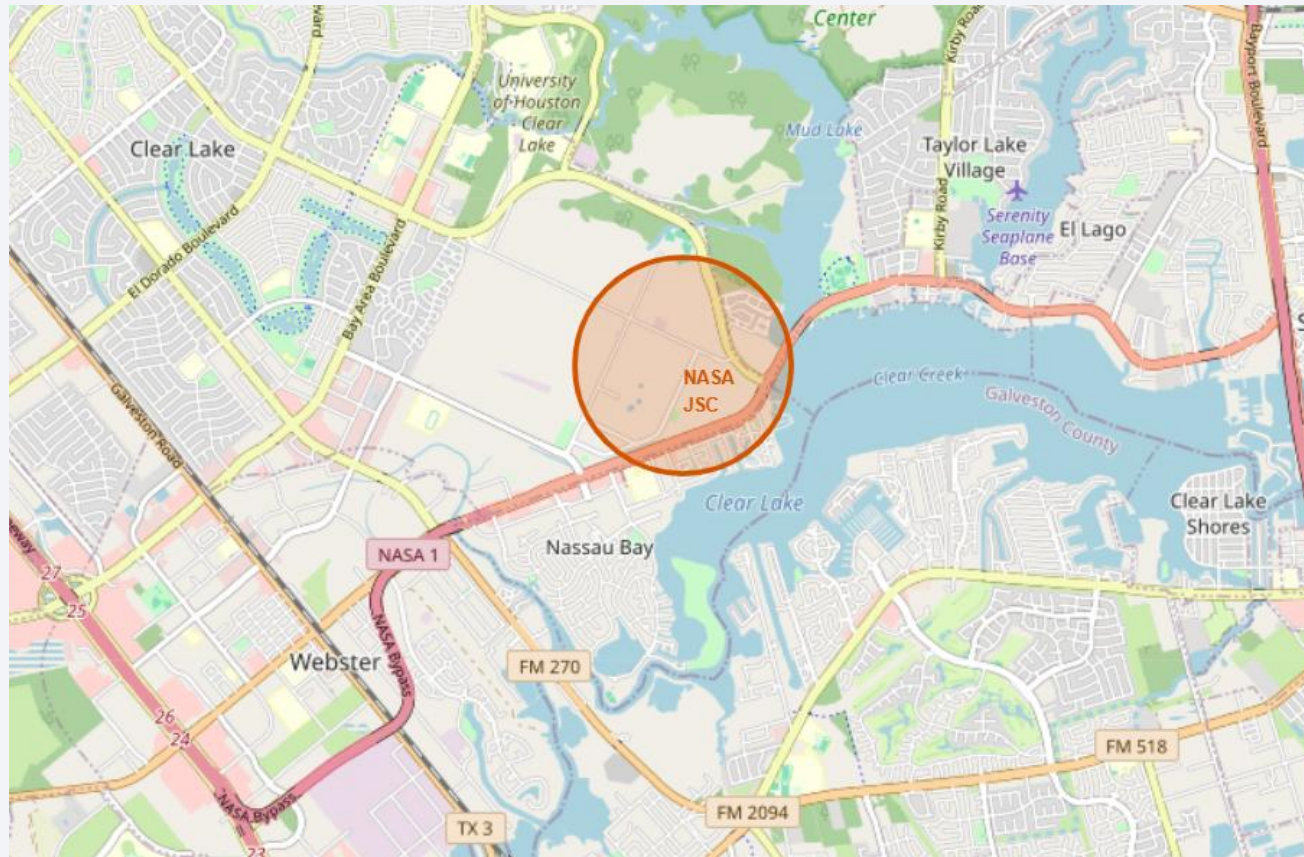
Query uses COUNT, WHERE, BETWEEN and GROUP BY

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

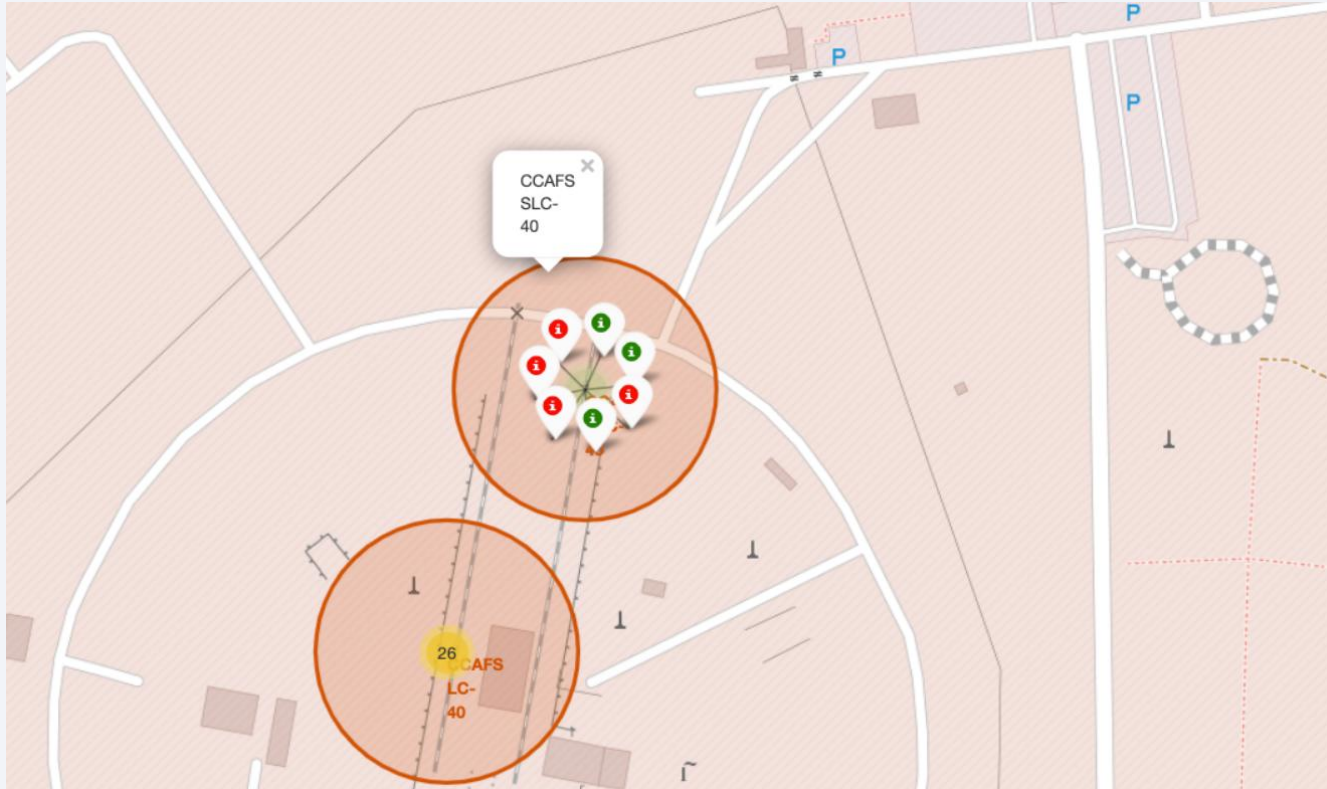
Launch Sites Proximities Analysis

Circle over the NASA



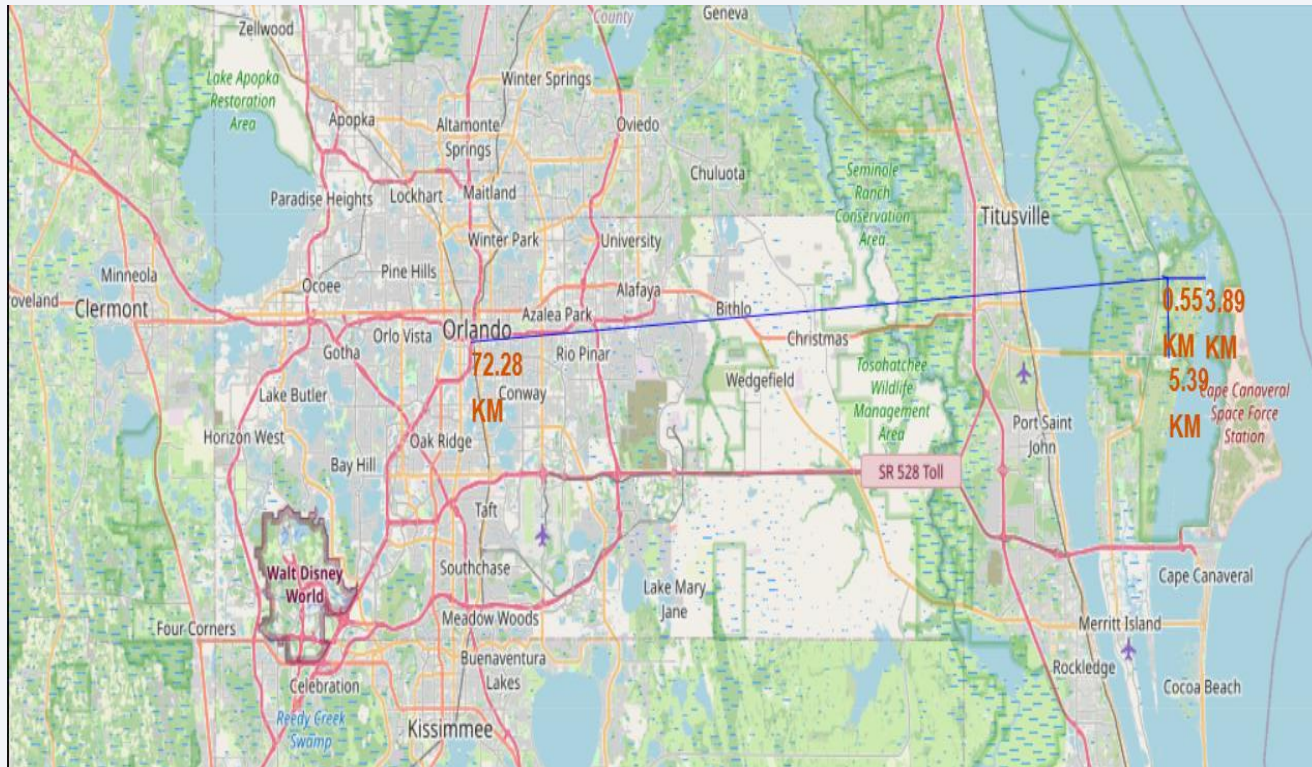
Map showing a circle over the NASA area near Houston.

Success Map

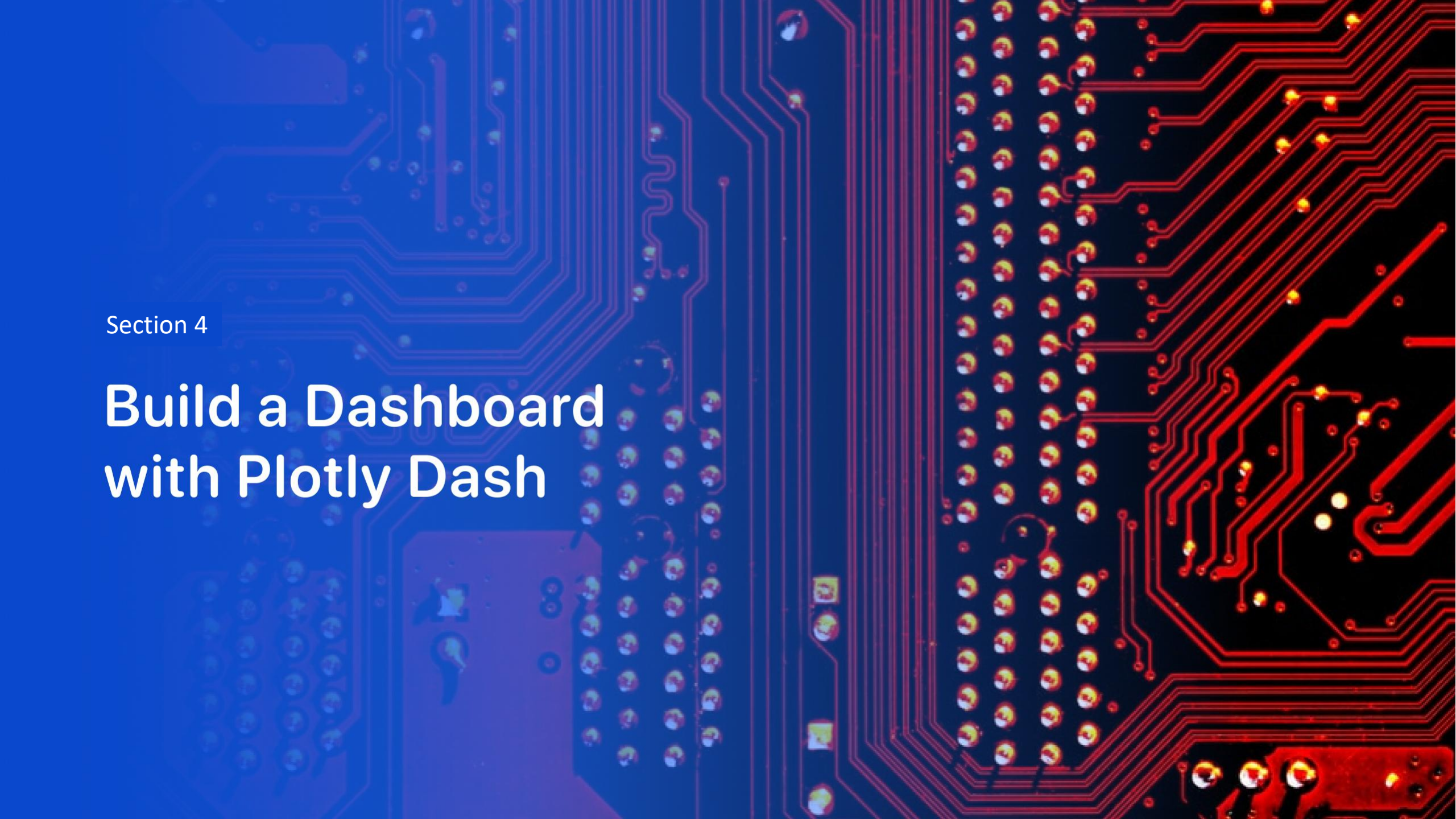


The map shows success in green and failures in red.

Points of interest



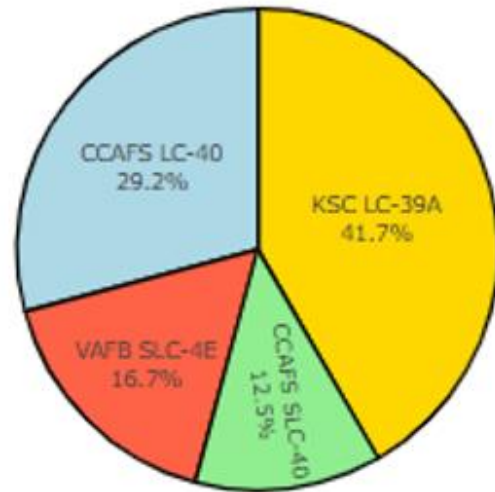
The map shows the distance of points of interest from Cape Canaveral.



Section 4

Build a Dashboard with Plotly Dash

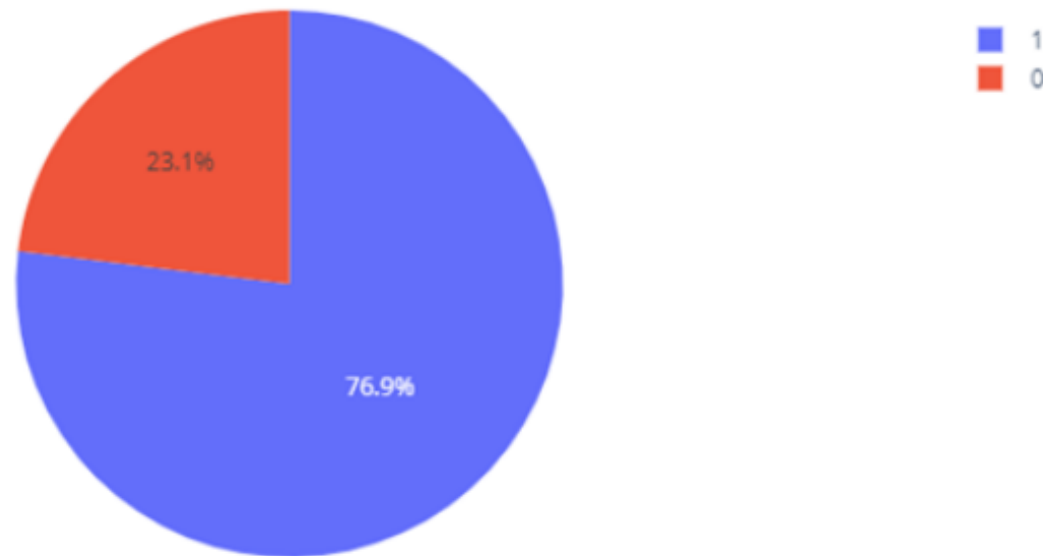
SPACEX – Total Success by Site



KSC LC 39 A is
the site with
high success

Success Launches - Sete KSC LC 39 A

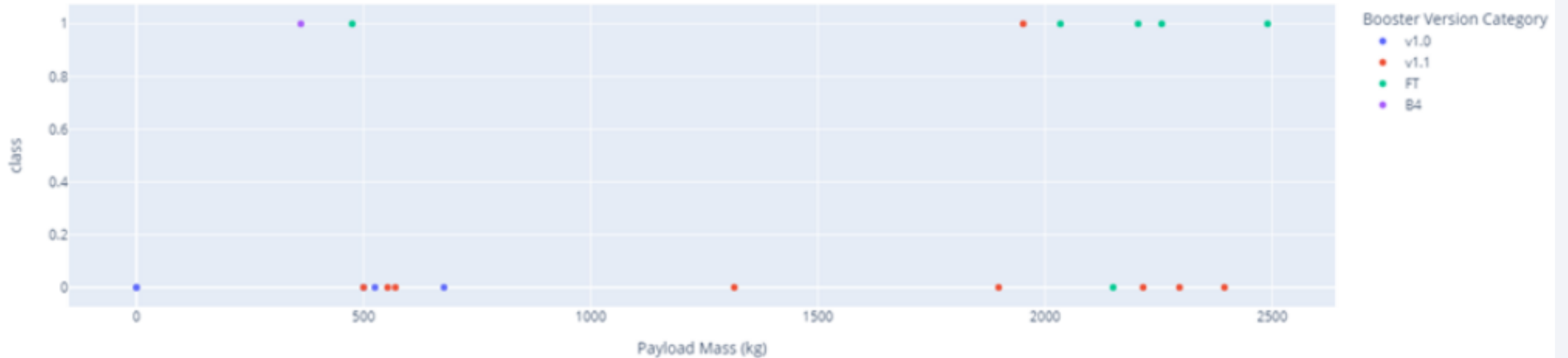
Total Success Launches for site KSC LC-39A



High
success
ratio

Payload x Launch outcome

Correlation between Payload and Success for all Sites

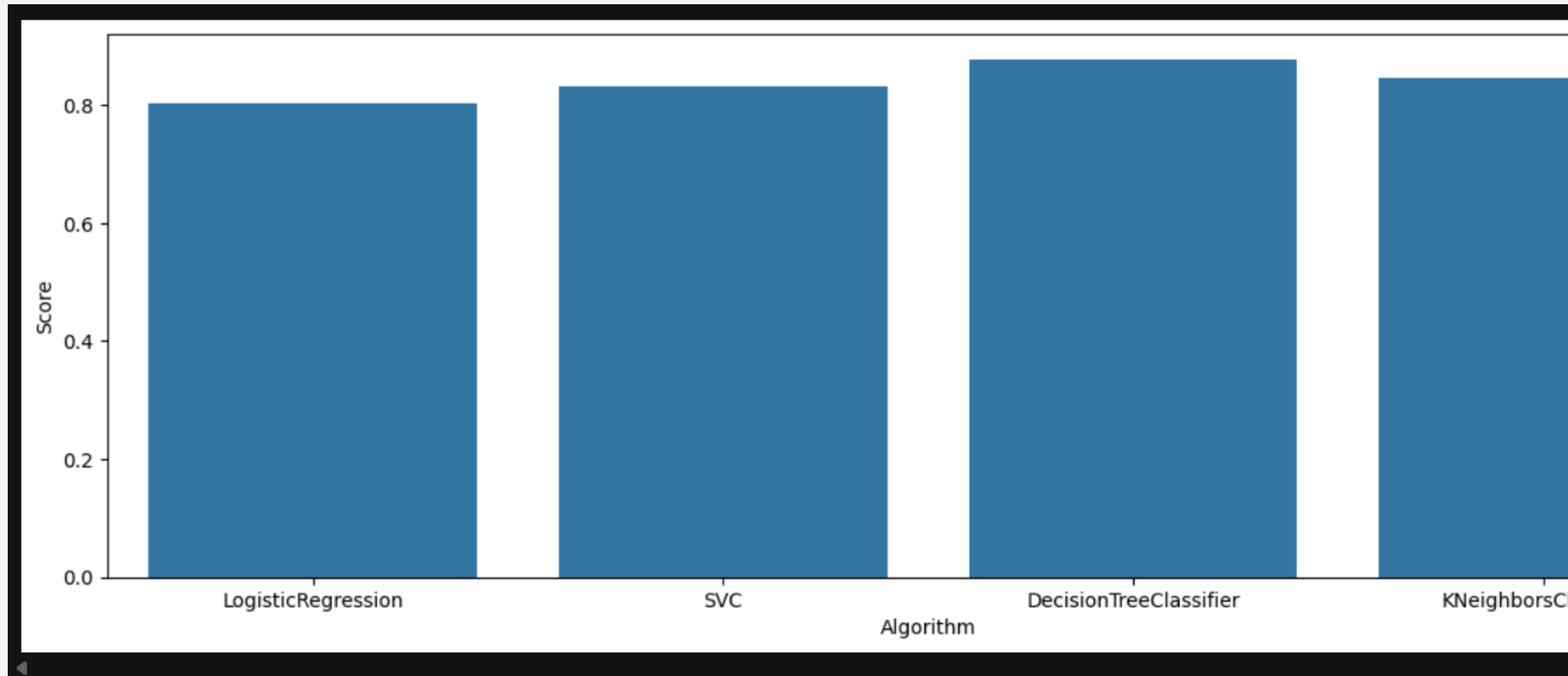


The 2500 – 5000kg range concentrate the major of successfully launch
The 0-2500Kg range concentrate the most failed launches

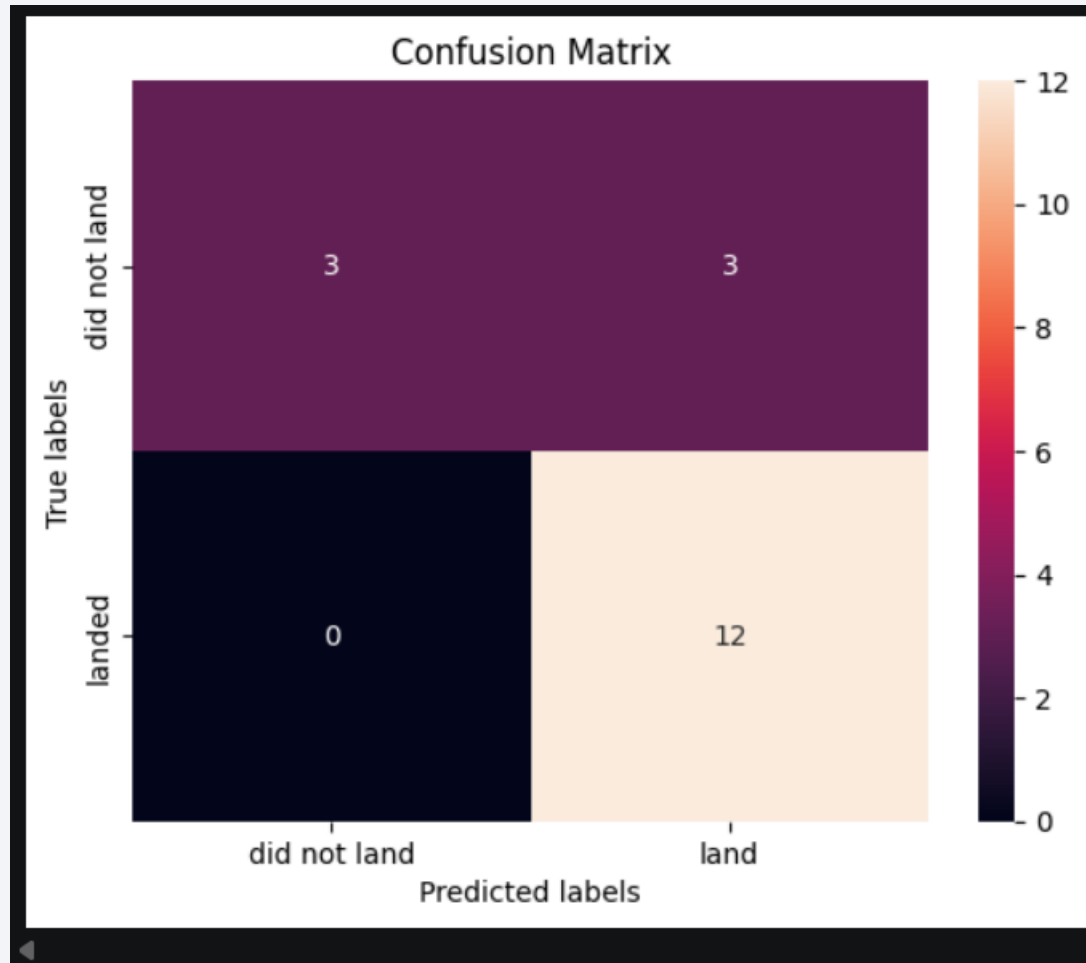
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



Conclusions

The Pipeline Worked End to End

From API collection and web scraping through cleaning, transformation and machine learning, every stage ran as an integrated workflow that turned unstructured real-world data into decision-ready insight.

Critical Success Factors Identified

Payload mass is the primary obstacle - heavier payloads sharply reduce landing odds.

Orbit type sets the difficulty - GTO is the most challenging, while LEO and GEO perform best.

KSC LC-39A outperforms the other sites thanks to its position closer to the Equator.

Success climbed from low levels in 2015 to above 80% in recent years as SpaceX iterated.

Predictive Modeling Delivered Robust Results

The four classifiers all exceeded 84% accuracy, with the Decision Tree selected as the final model at roughly 87.7%. Historical data carries more weight than real-time data, reducing risk before launch.

Practical Value

The results are directly applicable in the real world, helping teams make decisions before committing to a launch.

Models help teams understand and manage risk, lowering price and de-risk a mission.

Conclusions

What the SpaceX capstone project delivered

Falcon 9 first-stage landings can be predicted before launch with about 87.7% accuracy.

FINAL MODEL

Best accuracy and the strongest balance between precision and generalization.

KEY DRIVERS

Payload mass, orbit type and launch location shape the odds of a successful landing.

VALUE DELIVERED

Mission risk can be estimated before launch, saving cost and improving safety

Appendix

- SpaceX REST API - <https://api.spacexdata.com/v4/launches/past>
- Wikipedia web scraping - Falcon 9 and Falcon Heavy launch tables
- CSV datasets - spacex_launch_geo.csv, dataset_part_2.csv, dataset_part_3.csv
- [Github: marcelocards/capstone_spacex: Trabalho Capstone](#)

Thank you!

